YouTube Engineering Chatbot

RAG-Powered Q&A Bot for Mechanical Engineering Videos

65
Videos

1,400+
Chunks

3
Agent Tools

LangChain | Pinecone | GPT-4o-mini | Gradio | LangSmith

- Rishi

The Problem & Our Solution

😩 The Problem

- Hours of expert engineering content locked inside YouTube videos
- Scrubbing through videos to find a specific answer is time-consuming
- Students/People waste time rewatching lectures/videos
- Youtube search is keyword based not semantic

🚀 Our Solution

A conversational RAG chatbot that ingests entire YouTube channels, stores transcripts in a vector database, and answers precise questions with timestamps and source citations.

65	1,400 +	3
Videos Indexed	Transcript Chunks	Agent Tools

- Accessibility for hearing-impaired users
- Instant answers from hours of video content
- Source-cited responses build user trust

System Architecture — End-to-End Pipeline

1 — YouTube URLs
yt-dlp
youtube-transcript-api

→

2 — Text Processing
LangChain
RecursiveTextSplitter
700 chars / 150 overlap

→

3 — Quality Gate
Schema checks
Dedupe
PASS/WARN/FAIL

→

4 — Pinecone DB
text-embedding
-3-small
Cosine similarity

→

5 — RAG Agent/ Validation
GPT-4o-mini
3 tools
MemorySaver

→

6 — Gradio UI
Local deploy
Thumbnails+links

DATA FLOW

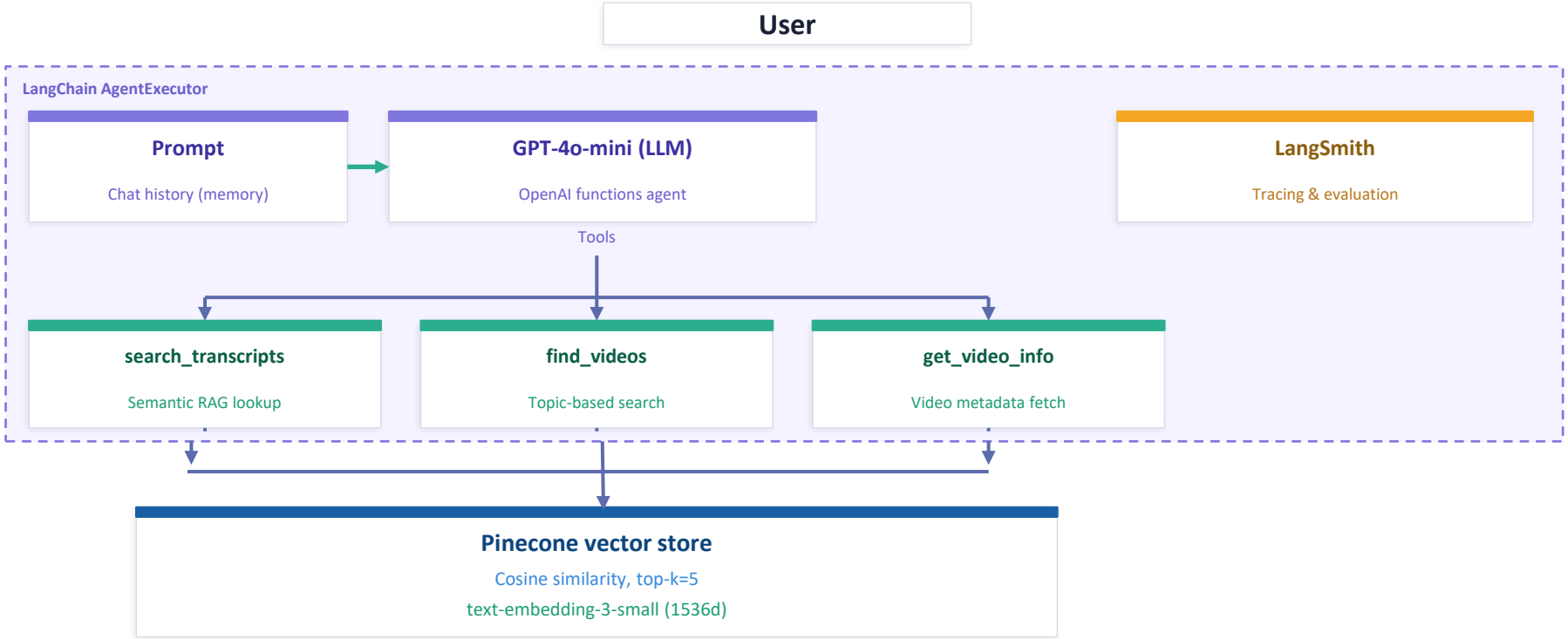
Ingestion Details

- Dataset: Youtube channels Efficient Engineer, Real engineering, NPTEL lectures on FEM
- Manual transcript preferred, auto-generated fallback
- JSON output: metadata + transcript + timestamps
- Supports single video, Shorts, and full playlists

Chunking Configuration

Chunk Size	700 chars
Overlap	150 chars
Min Chunk size	200 chars
Splitter	RecursiveCharacter
Timestamps	MM:SS per chunk

RAG Agent — System Architecture



LLM-as-judge evaluation

Accuracy · Hallucination · Relevance · Helpfulness

The Q&A Agent

LangChain AgentExecutor

GPT-4o-mini • OpenAI Functions

Ssession history— Manual memory

Memory – List of Human message / Ai message objects
- only available for current session

Tools

search_transcripts(query)

Core RAG tool — cosine similarity search returns top-K chunks with video title + timestamp

get_video_info(video_id)

Returns full metadata: title, channel, URL, duration, upload date for a given video

find_videos(topic)

Searches the index for videos covering a topic — deduped list of titles + watch URLs

Prompts

System Prompt Guardrails

No Hallucination

Answer ONLY from transcript evidence — never invent facts

Always Cite Sources

Every answer includes: video title + exact timestamp

Suggest Related

Recommend relevant videos when the topic allows

Admit Uncertainty

Clearly state when transcripts don't contain enough info

Evaluation: LangSmith LLM-as-Judge

Evaluation Method

LangSmith LLM-as-judge framework using GPT-4o to score agent answers against reference answers on a 1–5 scale.

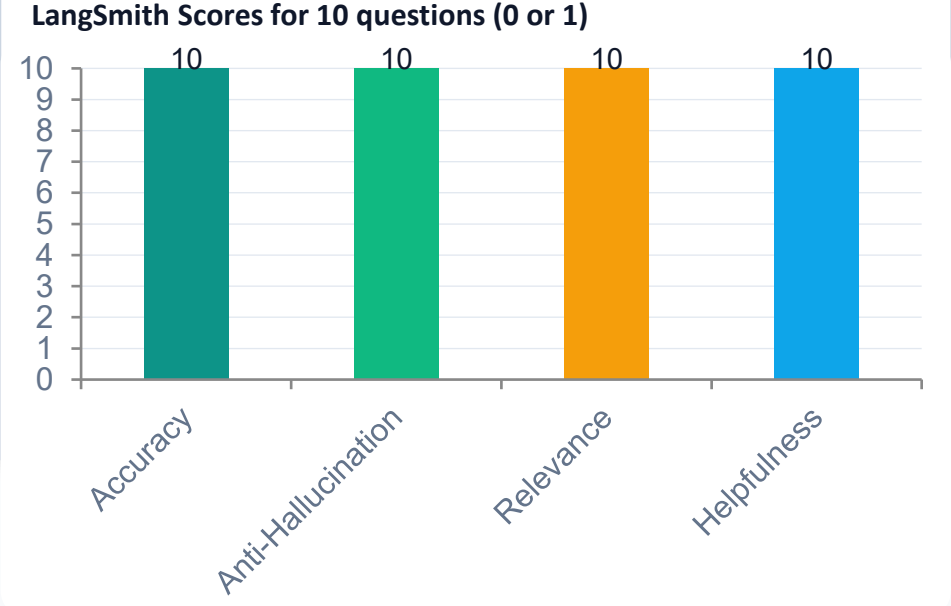
4 Dimensions Scored

Accuracy
Factually correct from transcripts

Anti-Hallucination
No invented facts outside transcripts

Relevance
Answer addresses the asked question

Helpfulness
Response is actionable and clear



Improvements after evaluation were due to

Better and stricter system prompts

Improving chunking configuration

Conclusions and leainings

What We Learned

Chunk quality > chunk quantity

Poor quality chunks in large quantities cannot yield good retrieval results

Timestamps unlock user trust

Users can verify answers at the exact video moment — a key differentiator over generic chatbots.

System prompts are guardrails

Without strict grounding instructions, LLMs confidently hallucinate engineering facts.

Model choice is a tradeoff between quality and cost

Future Roadmap

Voice Input (Whisper STT)

High Impact

Users speak their questions — Whisper transcribes to text for the agent.

Semantic Cache (FISE)

Performance

Cache frequent queries to reduce API costs and improve response latency.

Multi-Channel Expansion

Scale

Index multiple engineering channels for broader topic coverage.

Persistent Memory

UX

Database-backed cross-session chat history for returning users.

Live Demo

Deployed on Gradio site locally

http://127.0.0.1:7860

Pre-written Demo Script

1. ***"What is Turbulence?"***
   Shows core RAG + citation

2. ***" Can you tell me more about turbulence?"***
   Tests multi-turn memory

3. ***" Which videos cover fatigue failure?"***
   Demonstrates find videos tool

Thank you for you attention

Questions ?